

CYBERSECURITY INTERNSHIP PROGRAM

TASK - 1

Threat Intelligence Report

Modern Cybersecurity Threats (2024–2025) and Defense Measures

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1. Introduction to Cybersecurity

1.1 What is Cybersecurity?

Cybersecurity is the practice of protecting computer systems, networks, applications, and data from unauthorized access, damage, disruption, or theft. It spans a wide range of disciplines, including network security, application security, endpoint protection, cloud security, and incident response, all working together to preserve the confidentiality, integrity, and availability of information.

1.2 Why is it Important for Individuals and Businesses?

For individuals, cybersecurity protects personal data, financial accounts, identity information, and privacy from criminals seeking to exploit them for fraud or extortion. For businesses, it protects intellectual property, customer data, financial assets, and operational continuity. A single successful attack can lead to direct financial loss, regulatory penalties, legal liability, and lasting reputational damage that affects customer trust and shareholder confidence.

1.3 Current Relevance

Cybercrime has grown sharply in scale and sophistication. The FBI's Internet Crime Complaint Center (IC3) recorded roughly \$16.6 billion in reported cybercrime losses in 2024, a 33% increase over the previous year. This growth is driven by three converging factors: rapidly increasing digital dependency across personal and business life, the expanding attack surface created by cloud computing and connected (IoT) devices, and the emergence of generative AI, which is now being weaponized by attackers to create convincing phishing content, cloned voices, and deepfake video impersonations at a scale and quality that was not possible even two years ago.

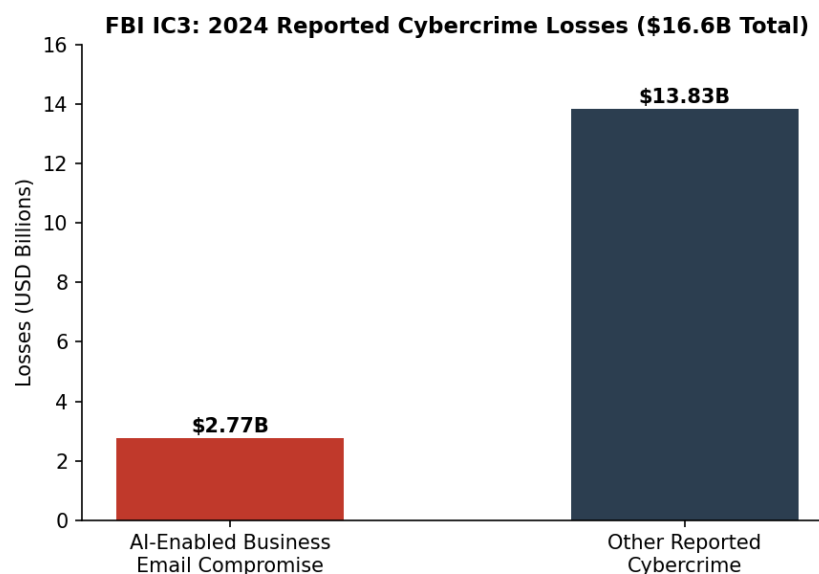


Figure 1: AI-enabled Business Email Compromise accounted for roughly one-sixth of all reported 2024 cybercrime losses in the United States. Source: FBI IC3 / Vectra AI.

2 & 3. Major Modern Cyber Threats, Impact Analysis, and Case Studies

The five threats below were selected because they represent the categories most frequently cited by CISA, IBM Security, and independent researchers as dominating the 2024–2025 threat landscape. Each includes an impact analysis, a documented real-world incident, and recommended defenses.

2.1. AI-Powered Phishing Attacks

Generative AI has removed most of the traditional warning signs of phishing — poor grammar, generic messaging, and unnatural phrasing. Attackers now use AI to write fluent, personalized spear-phishing emails, clone voices from short audio samples, and generate real-time deepfake video during calls, making impersonation of executives or colleagues highly convincing.

Impact Analysis

Affected Party	Impact
Individuals	Individuals can be tricked into revealing passwords, banking details, or one-time codes, leading to direct financial theft and identity fraud. Victims may also suffer emotional distress and long recovery times reclaiming compromised accounts.
Organizations	Organizations face large, fast financial losses through fraudulent wire transfers, plus follow-on costs from incident response, legal exposure, and reputational damage. Because these attacks bypass traditional red-flag training, employee trust in normal video/voice communication is also undermined.

Real-World Case Study

In early 2024, a finance employee at the Hong Kong office of British engineering firm Arup received what appeared to be a routine request from the company's CFO. He initially suspected phishing, but was then invited to a video conference where he saw and heard the CFO and several colleagues — all of whom were AI-generated deepfakes. Believing the meeting was genuine, he authorized transfers totaling roughly \$25.6 million to the attackers across multiple transactions. The case is one of the largest publicly confirmed deepfake fraud incidents to date and has been widely cited by CNBC, the FBI's IC3, and multiple security research teams as a turning point for AI-enabled fraud.

Preventive Measures

- Require independent, out-of-band verification (a callback to a known number, or a pre-agreed code word) for any request to transfer funds or credentials, even when a call or video looks legitimate.
- Update security awareness training to cover voice cloning and deepfake video, not just email red flags.
- Deploy AI-based detection tools and enforce multi-layered approval for high-value transactions.

2.2. Ransomware-as-a-Service (RaaS)

RaaS is a criminal business model in which ransomware developers lease their malware, infrastructure, and negotiation support to affiliates in exchange for a share of ransom payments. This has dramatically lowered the technical skill needed to launch a ransomware attack, and double-extortion (encrypting data and threatening to leak it) has become standard practice.

Impact Analysis

Affected Party	Impact
Individuals	Individuals whose personal data is held by a breached organization face identity theft, financial fraud, and long-term exposure if stolen data is leaked or sold on criminal marketplaces.
Organizations	Organizations suffer operational downtime (sometimes for weeks), direct ransom costs, data loss, regulatory fines, and reputational damage. Critical-sector victims (healthcare, government, logistics) risk service disruption with public safety consequences.

Real-World Case Study

LockBit became the world's most prolific RaaS operation, linked to as much as 30% of all ransomware attacks before international law enforcement (the UK's National Crime Agency, FBI, and Europol) seized parts of its infrastructure in February 2024 under “Operation Cronos.” Despite the takedown, the group resurfaced within days and continued operations through 2024 with a new variant. Industry data shows the average ransomware claim in the first half of 2024 exceeded \$5.2 million, including one record payment of \$75 million, illustrating how RaaS has scaled criminal ransomware into an industrialized business model.

Preventive Measures

- Maintain regular, offline (air-gapped) backups and routinely test data restoration.
- Deploy Endpoint Detection and Response (EDR) and segment networks to limit lateral movement.
- Apply rigorous patch management and maintain a tested incident response plan for rapid containment.

2.3. Cloud Security Misconfigurations

As organizations move infrastructure to AWS, Azure, and GCP, human error in configuring storage, identity, and access permissions has become one of the leading causes of data breaches. Publicly exposed storage buckets, overly permissive identity and access management (IAM) policies, and missing encryption are among the most common mistakes.

Impact Analysis

Affected Party	Impact
Individuals	Individuals whose personal data is stored by an affected organization face exposure of names, contact details, financial information, or credentials, increasing the risk of fraud and follow-on phishing.
Organizations	Organizations face regulatory penalties (e.g., under GDPR or similar data protection laws), loss of customer trust, legal liability, and in some cases direct theft of source code or intellectual property alongside customer data.

Real-World Case Study

In 2024, Football Australia suffered a data breach after developers misconfigured the organization's AWS S3 storage buckets, leaving them publicly accessible. The exposed data included attendee purchase information, internal computing design documents, and source code. Investigators classified the root cause as a failure of configuration and change control — the same category of error responsible for a large share of all cloud breaches industry-wide; cloud misconfigurations were estimated to have caused over \$6 billion in losses in 2024 alone.

Preventive Measures

- Adopt a Cloud Security Posture Management (CSPM) tool to continuously scan for misconfigurations.
- Enforce least-privilege IAM policies and block public access on storage services by default.
- Enable encryption at rest and in transit, and conduct regular configuration audits and change-control reviews.

2.4. IoT Vulnerabilities

Internet of Things devices — routers, IP cameras, DVRs, smart home devices — are frequently shipped with weak default credentials, limited built-in security, and infrequent firmware updates. This makes them attractive, low-effort targets for attackers building large botnets.

Impact Analysis

Affected Party	Impact
Individuals	Individuals face loss of privacy (compromised cameras), disrupted home networks, and can unknowingly have their devices used as launch points for attacks on others.
Organizations	Organizations relying on IoT devices for operations (security cameras, industrial sensors) face operational disruption, and businesses hosting services can suffer outages when targeted by IoT-botnet-driven Distributed Denial of Service (DDoS) attacks.

Real-World Case Study

Beginning in mid-2024, threat actors exploited an unpatched vulnerability in Edimax IP security cameras — commonly used in commercial and industrial settings — using default login credentials to install Mirai botnet malware. With no patch available, affected organizations had to replace hardware entirely. Separately, a Mirai-based botnet exploiting a command-injection flaw (CVE-2024-3721) in TBK DVR devices was used to launch large-scale DDoS attacks against the maritime logistics sector, and by mid-2024 a related nation-state-linked IoT botnet was found to have compromised over 260,000 devices worldwide.

Preventive Measures

- Change default usernames and passwords immediately upon device setup.
- Apply firmware and security updates as soon as manufacturers release them.
- Segment IoT devices onto a separate network (VLAN) isolated from critical business systems.

2.5. Zero-Day Exploits

A zero-day is a vulnerability that is exploited by attackers before the vendor is aware of it and before a patch exists. Because no defense signature or patch is available at the time of attack, zero-days are especially valuable to sophisticated and state-sponsored threat actors targeting high-value enterprise systems.

Impact Analysis

Affected Party	Impact
Individuals	Individuals are typically affected indirectly — through compromise of a service or organization that holds their data — rather than being targeted directly.

Organizations	Organizations face potentially total network compromise, since zero-days are frequently used to gain an initial foothold before attackers move laterally, exfiltrate data, or deploy ransomware. Recovery often requires full forensic investigation and infrastructure rebuilding.
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Real-World Case Study

In January 2024, security researchers disclosed two zero-day vulnerabilities (CVE-2023-46805 and CVE-2024-21887) in Ivanti Connect Secure and Policy Secure VPN appliances, which had already been actively exploited in the wild since December 2023 by a suspected China-linked espionage group. The flaws allowed authentication bypass and command injection, enabling full unauthenticated remote access. The impact was severe enough that the U.S. Cybersecurity and Infrastructure Security Agency (CISA) issued an emergency directive requiring all federal agencies to disconnect affected appliances by February 2, 2024. Google's Threat Intelligence teams tracked 75 zero-day vulnerabilities exploited in the wild during 2024 overall, with 44% targeting enterprise products directly.

Preventive Measures

- Maintain a rapid patch management process and subscribe to vendor and CISA security advisories.
- Deploy Intrusion Detection/Prevention Systems (IDS/IPS) and network monitoring to catch anomalous behavior even without a known signature.
- Adopt a Zero Trust security model so that a single compromised system does not grant broad network access.

6. Conclusion & Future Scope

The threats profiled in this report — AI-powered phishing, Ransomware-as-a-Service, cloud misconfigurations, IoT vulnerabilities, and zero-day exploits — share a common thread: attackers are professionalizing, automating, and scaling their operations faster than many organizations are scaling their defenses. RaaS has turned ransomware into a service industry; generative AI has made social engineering nearly indistinguishable from genuine communication; and the sheer growth of cloud and IoT infrastructure has expanded the attack surface faster than security teams can audit it.

Proactive cybersecurity — patching before exploitation, verifying before trusting, and monitoring continuously rather than reactively — is no longer optional for individuals or organizations operating in a digitally dependent economy. Equally important is continuous learning: the specific threats will keep changing as attackers adopt new technology, so the underlying discipline of staying current, testing defenses, and questioning unusual requests is what will remain valuable regardless of which threat is dominant next year.

Looking ahead, AI is likely to play a growing role on both sides of the fight — as a tool for attackers generating more convincing deception, and as a tool for defenders detecting anomalies at a scale humans cannot match. Organizations that invest now in Zero Trust architectures, cloud security posture management, and workforce training that reflects current (not outdated) threats will be best positioned as the landscape continues to evolve.

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